



Who Is Ahead?

Run both, then compare the landings

Name:

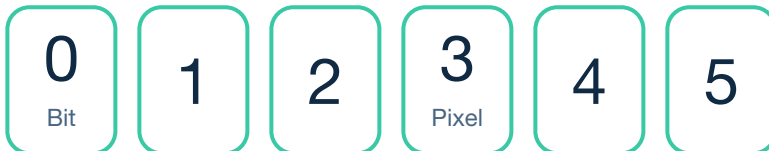
Date:

★ I can run two scripts at the same time and compare where each sprite lands.

Higher number = ahead

Bit (green) and Pixel (purple) both start on the green flag ► and run at the SAME time. On the number path, a HIGHER number means a sprite is further ahead. Read each script, find where each lands, then compare: who is ahead?

The number path



Bit

Bit's script

► start at 0

forward 4

back 1

1 Where does Bit land?

Read Bit's script from 0: forward 4, then back 1. What number does Bit land on?



Pixel

Pixel's script

▶ start at 3

forward 1

forward 1

2 Where does Pixel land?

Read Pixel's script from 3: forward 1, then forward 1. What number does Pixel land on?

3 Who is ahead?

The green flag starts BOTH scripts at once. Write where each lands, then circle who is AHEAD (higher number). Bit: ____ Pixel: _____. Who is ahead, and by how many numbers?

Big idea

Both scripts run at the same time, so we can compare them. The sprite on the higher number is ahead. Here you compare Bit's landing and Pixel's landing to see who is in front.



Who Is Ahead?

Quick facilitation notes & answer key

What this worksheet teaches

Students run both characters' moves and compare: who is on the higher number (ahead), and by how many. Comparing the two landings — not just reading one — is the Grade 2 step. No coding experience needed.

Before you start (no computer needed)

No computer needed — paper and a pencil. You read the prompts aloud. Optional: put two counters on a number line 0 to 5 — green for Bit, purple for Pixel — and move BOTH at the same time so children see the two sets of moves run side by side.

How to lead it (about 20 minutes)

1. Show them (you do). Run both sets of moves, then point out which character ended higher.
2. Do one together. Exercise 1 — Bit lands on 3; Exercise 2 — Pixel lands on 5.
3. Let them try. Students write both landings, circle who is ahead, and find by how many (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — Bit: 0, forward 4 → 4, back 1 → 3. Exercise 2 — Pixel: 3, forward 1 → 4, forward 1 → 5.

Exercise 3 — both start on the green flag; Pixel (5) is on the higher number, so Pixel is ahead, by $5 - 3 = 2$.

Key idea: the two sets of moves run at the same time; compare the final positions to decide who is ahead.

If students get stuck — and ways to adjust

Watch for: saying "ahead" by who finished first. Ahead means the higher number on the line.

Make it easier: line up the green and purple counters and see which is further right.

Make it harder: ask what move would change who is ahead.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

In plain terms: students write and run instructions, including two sets that run at the same time.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

In plain terms: students read two sets of instructions, change one, and explain what happens when both run together.

You'll know it worked when

The child finds both landings, names who is ahead, and by how many.